Biological and environmental data for SDM

Modeling the distribution of marine biodiversity under global climate change

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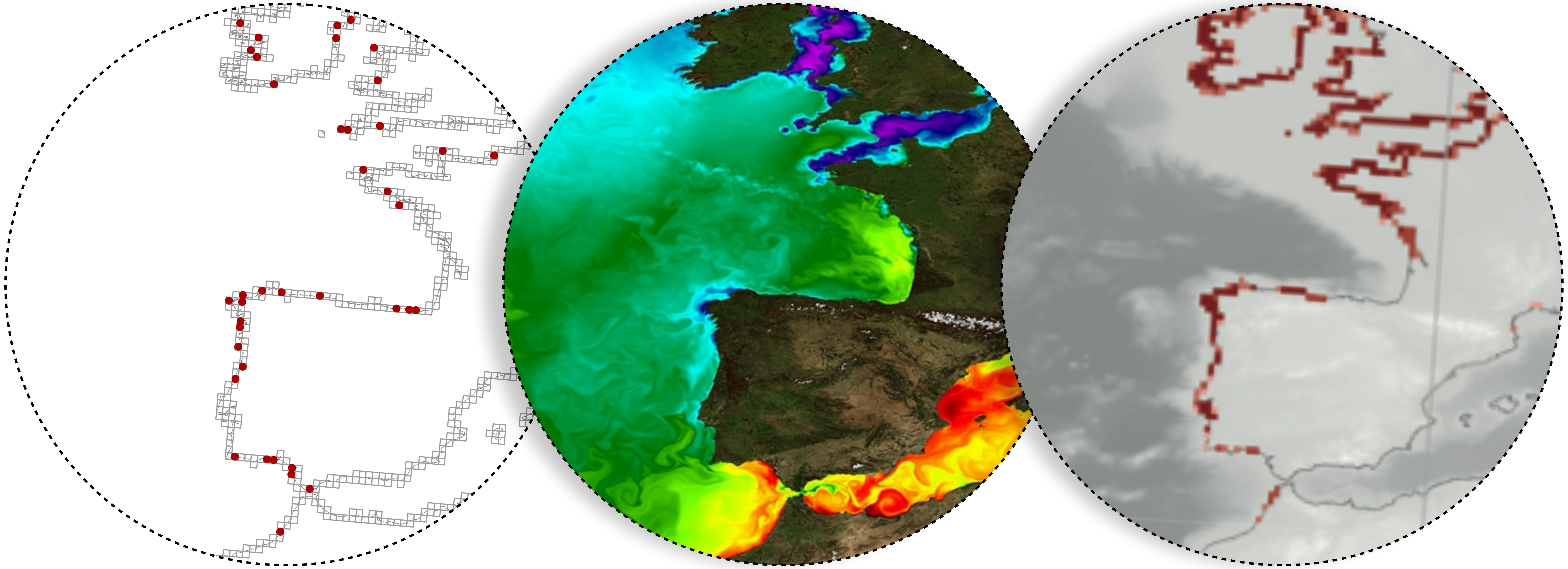
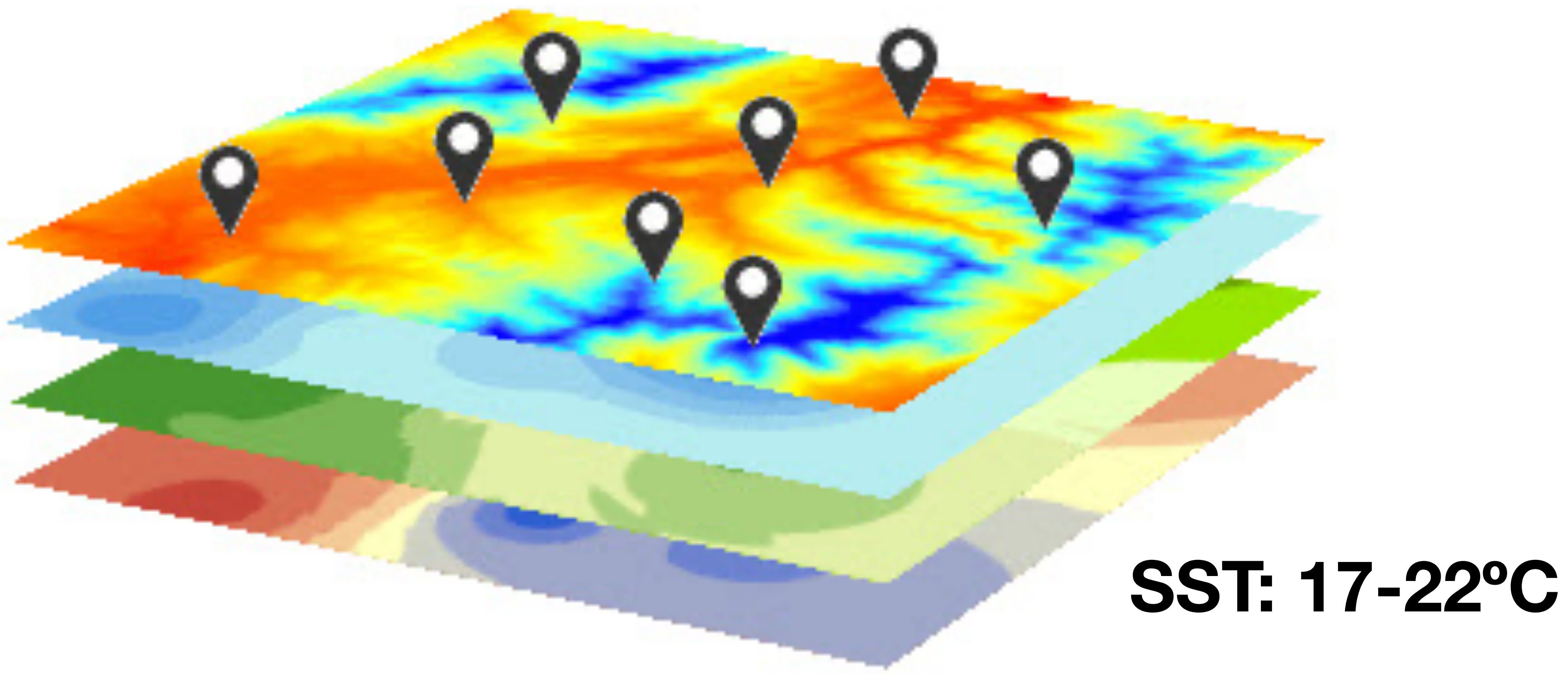


Occurrence records

Environmental layers

Distribution model

SDMs rely on species occurrence records (where are the species found) and environmental data layers (what are the conditions across the study area).



Occurrence records for modeling

Allow building an SDM and answer the question of **why a species occurs in a given area.**

Occurrence records need to reference, at least, a **particular taxon**, at a **particular place**, and a **particular time**.

Geographic coordinates are key, as they allow **linking occurrences records with environmental conditions at sites**.

Metadata can describe the source of record (sampling method), means of identification (e.g., visual census), and spatial precision of record (maximum error).

Occurrence data for SDM : one species example

Data

Metadata

Lon	Lat	Date	Depth	Species	Common name	Behaviour	Observer
0,61287	39,58955	10/08/12	22	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
0,83999	39,77417	03/03/12	12	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
3,18192	42,40479	14/08/11	1	Caretta caretta	Loggerhead sea turtle	Eating	Unknown
-4,86395	55,24295	01/01/04	0	Caretta caretta	Loggerhead sea turtle	Swimming	Unknown
-5,89297	57,06977	03/03/12	0	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
34,9082	32,55822	22/12/02	23	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
0,61287	39,58955	14/08/11	12	Caretta caretta	Loggerhead sea turtle	Swimming	John Doe
0,83999	39,77417	03/03/12	2	Caretta caretta	Loggerhead sea turtle	Sleeping	John Doe
3,18192	42,40479	22/12/02	5	Caretta caretta	Loggerhead sea turtle	Eating	Unknown
-4,86395	55,24295	03/03/12	7	Caretta caretta	Loggerhead sea turtle	Swimming	Unknown
-5,89297	57,06977	01/01/04	3	Caretta caretta	Loggerhead sea turtle	Eating	Unknown

Occurrence records: availability

Where is the data available for modelling?

- Global Biodiversity information centres;
- Literature;
- Reference collections (e.g. museums and herbaria);
- Citizen science initiatives.

Occurrence records: Global Biodiversity information centres

(i.e., databases linked via a common access portal).

Global Biodiversity Information Facility (GBIF):

Occurrence records: over 1.9 billion

Ocean Biodiversity Information System (OBIS):

Occurrence records: over 50 million

National Biodiversity Network (NBN) Atlas (UK):

Occurrence records: over 200 million

Atlas of Living Australia (ALA):

Occurrence records: over 95 million

Occurrence records: Biological reference collections

Preserved specimens / **photographs** maintained in **museums**, **universities**, **botanic gardens**, and similar **institutions**.

A major tools for basic investigation and assessment of biodiversity.

~2.5 billion specimens in preserved collections.

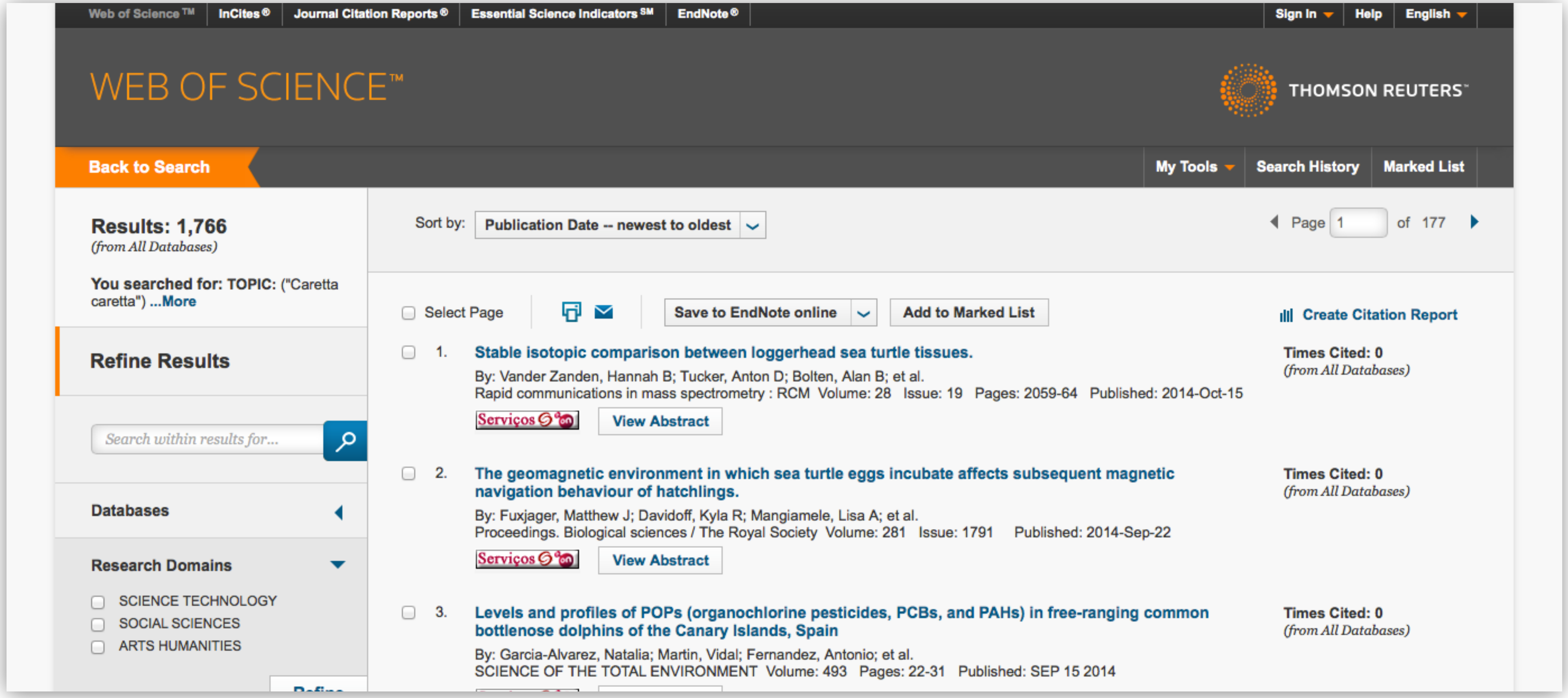
Macroalgae Herbarium Portal (www.macroalgae.org):

Occurrence records: 1 million records.

Occurrence records: Literature

Literature is useful to compile biodiversity records.

Peer-reviewed publications are preferred, even when out of the scope of our work.



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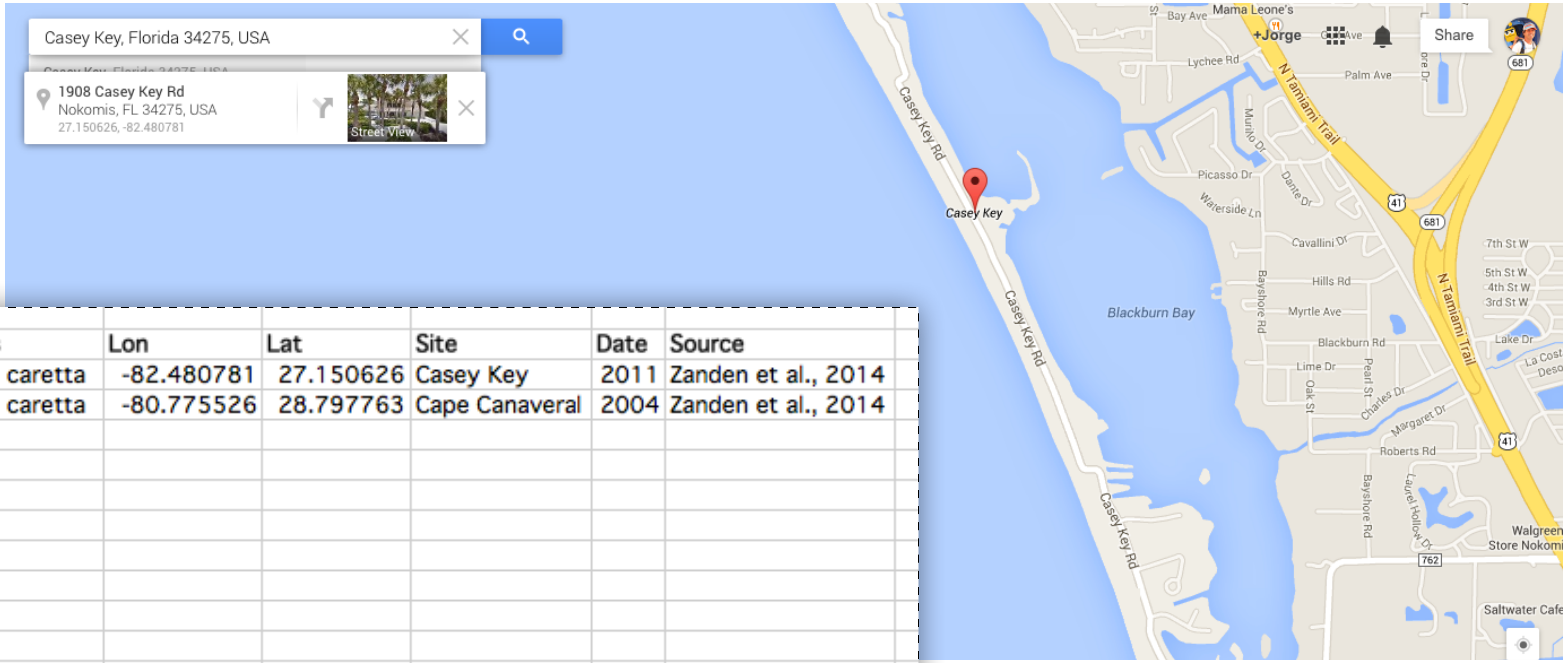
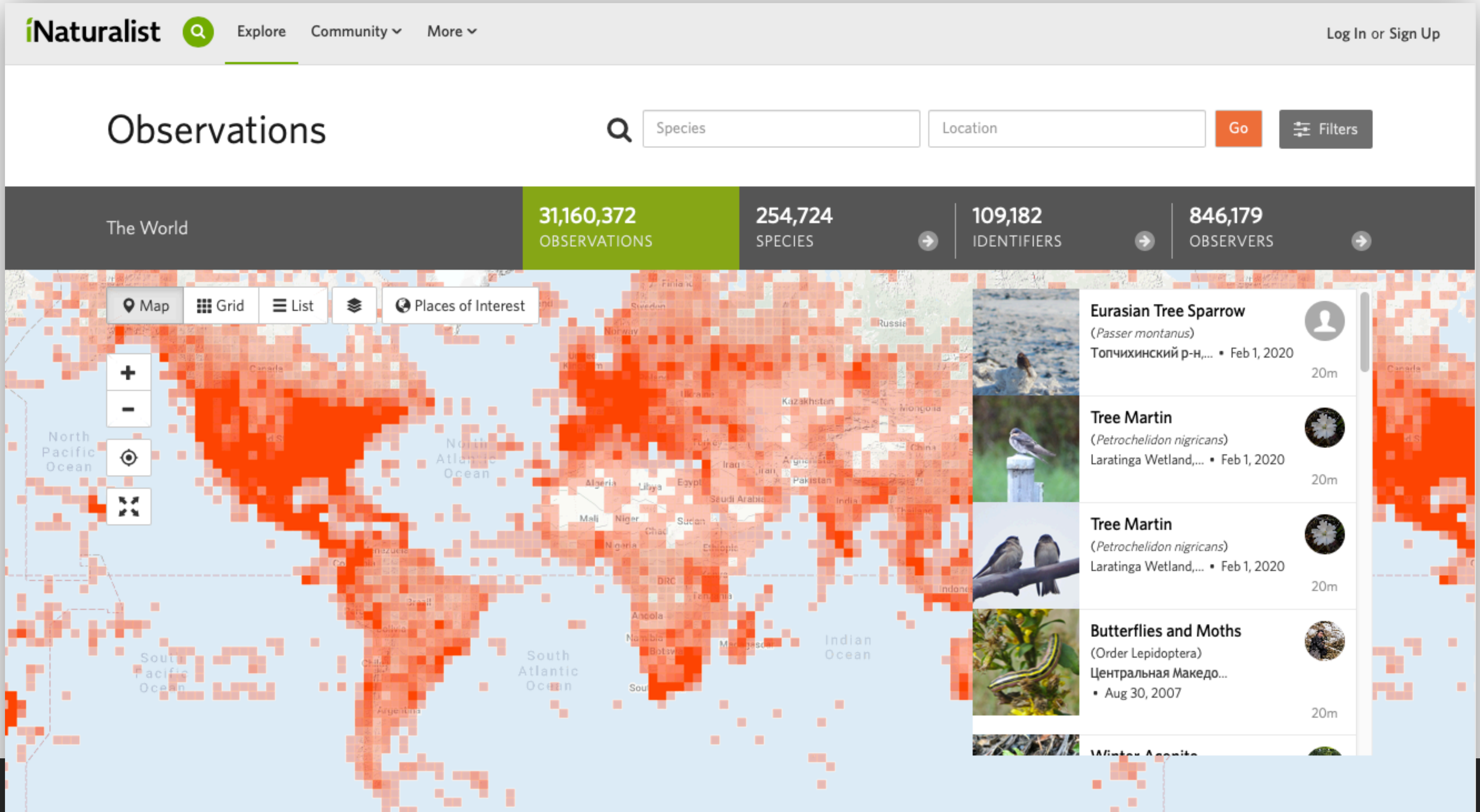


Species	Lon	Lat	Site	Date	Source
Caretta caretta	-82.480781	27.150626	Casey Key	2011	Zanden et al., 2014
Caretta caretta	-80.775526	28.797763	Cape Canaveral	2004	Zanden et al., 2014

Occurrence records: Citizens can help

Sharing biodiversity records / pictures with scientists.

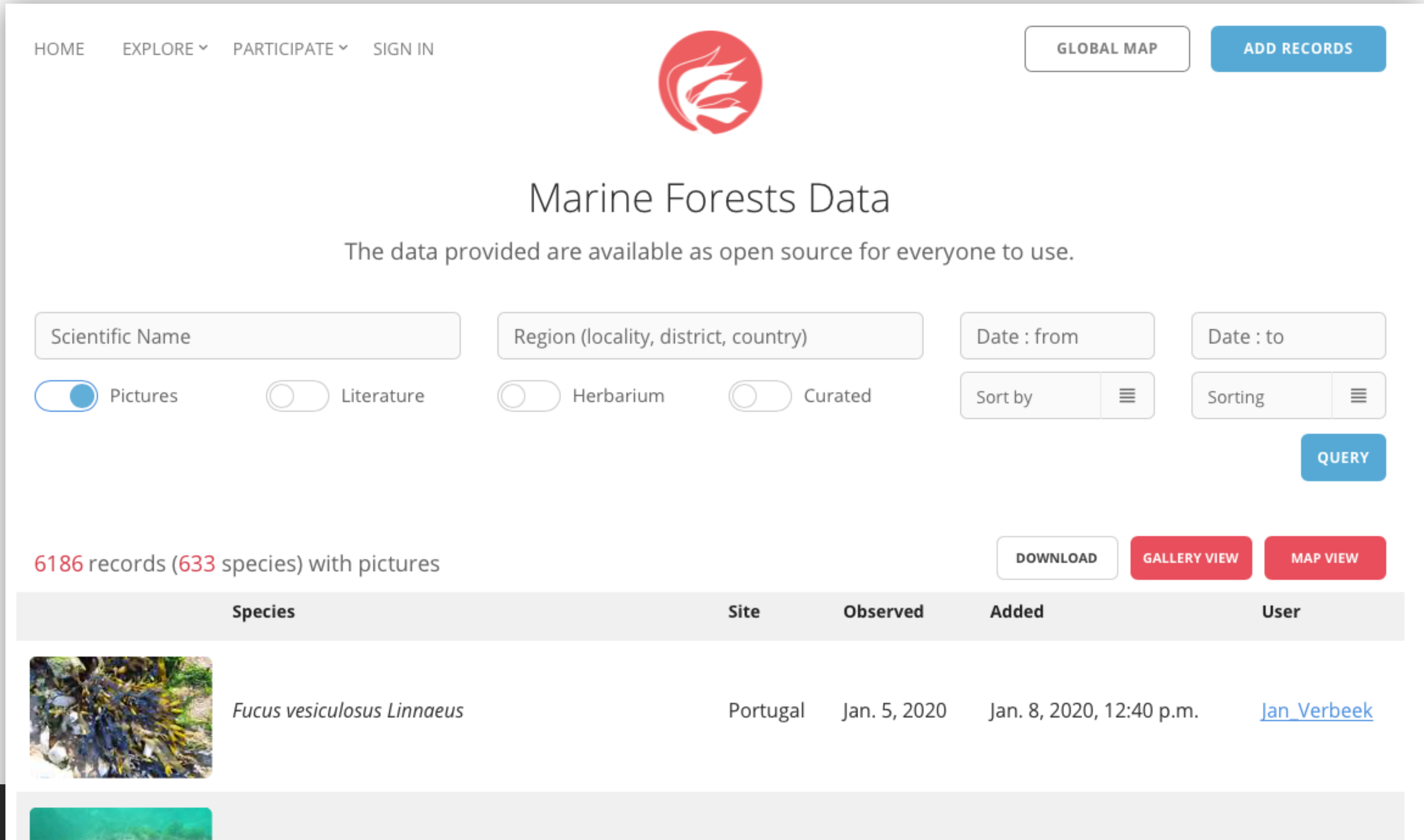
http://www.inaturalist.com



Occurrence records: Citizens can help

Sharing marine forest (kelps, seagrasses, etc.) pictures with scientists.

http://www.marineforests.com



Occurrence records: issues

“Garbage In, Garbage Out": model outputs are only as good as the input data.

The quality of several sources has been questioned.

There are **inaccuracies** and **bias** in occurrence records.

Inaccurate records

1. Inaccurate taxonomy of species

Manual **curation and** expert **judgment is needed** prior to modeling.

One must know the overall distribution of the modeled taxon to properly judge occurrence records.

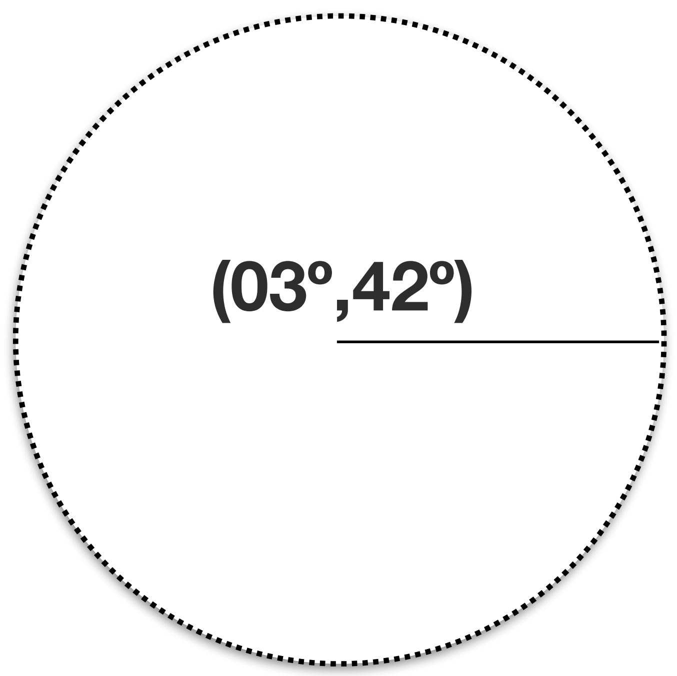
2. Inaccurate coordinates

Coarse resolution data has little meaning on where record were made.

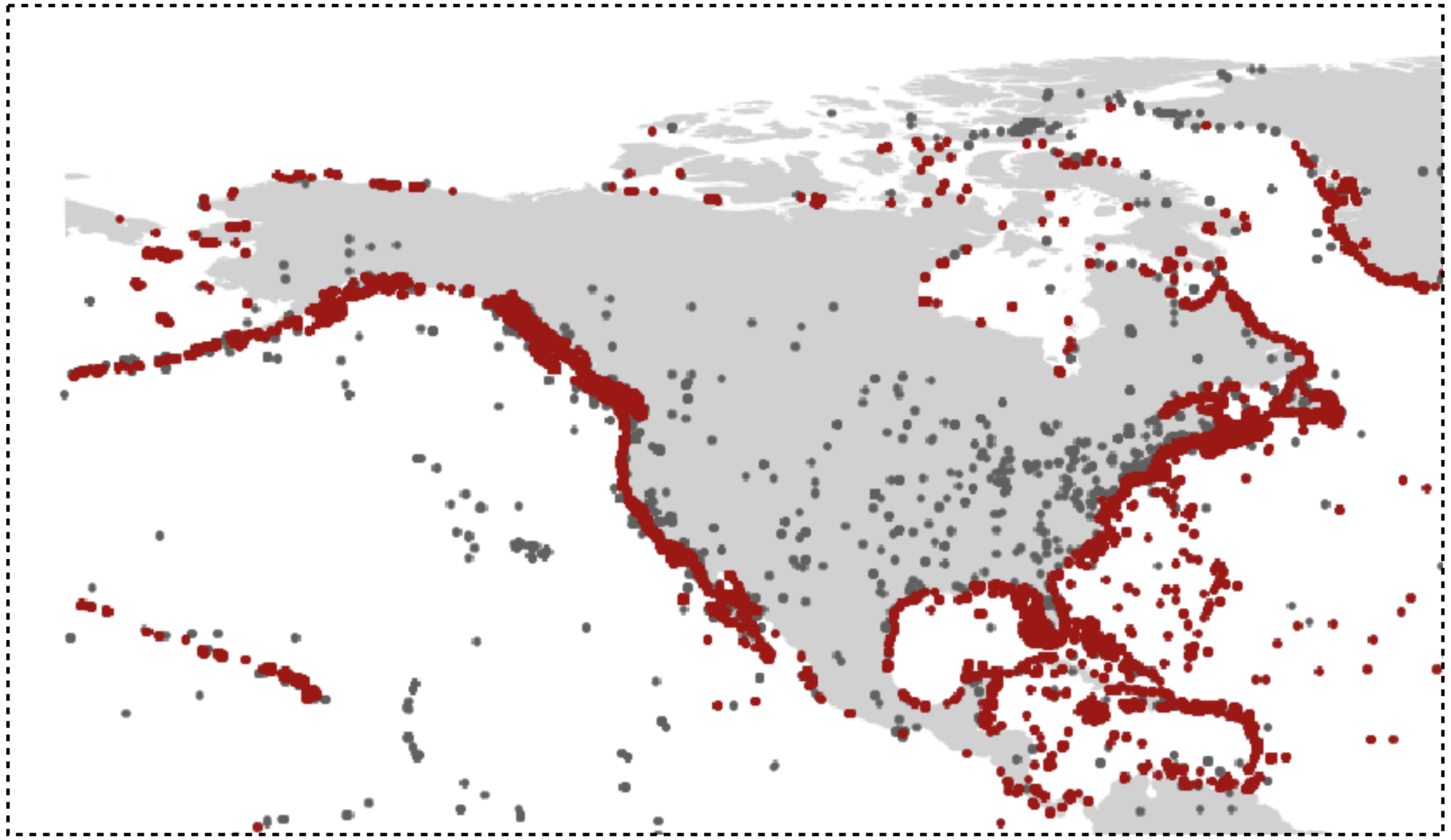
e.g.,

Natural history specimens may have very general **locality information**.

Brazil, Rio de Janeiro: locality descriptor refers to the **city** or the **state**? Also, in 2022 it is **a broader city** than it was in 1880. Also, how far outside of the limits of Rio de Janeiro the specimen could have been collected and still be labeled as “Rio de Janeiro”.



“Longitude: 3° and Latitude 42°”: A record with an accuracy at the scales of 1 degree has a potential error of ~111 kilometres.



Inaccurate records

Marine records over land (and vice versa)

Can also arise from coordinate imprecision, flipped Lon-Lat or wrong source information (the museum location instead of records).

Can be easily dealt with.

Biased records

Detectability

Some species more easily detected.

> The **more mobile or cryptic the species, the greater is the detection problem**.

Raises the change of **false absences** when characterizing biodiversity data, which can lead to wrong estimates of unsuitable habitats.

This sort of bias **rarefies biodiversity without introducing systematic biases, because the effect is consistent across range**.

Biased records

Distribution effort

Concentrate records, providing a skewed representation of biodiversity.

1. Sampling **near roadways**, **rivers**, areas of high human **population density** (common in terrestrial ecology).

2. The presence of a particular **institution** or **investigator in a given area**, or the location of a popular **field station**.

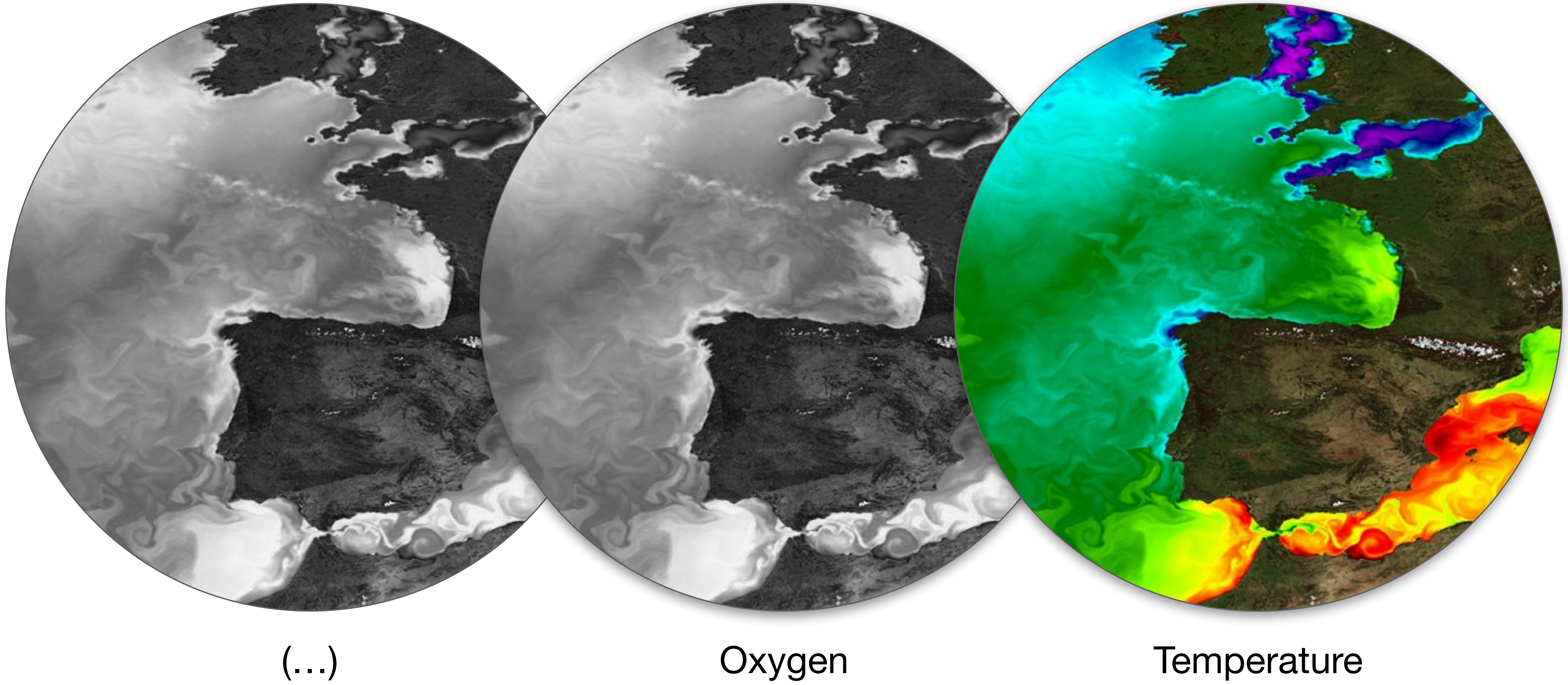
3. Particular organizations or **scientists** not **willing to share data**;

4. **Inaccessibility to particular regions, e.g., deep waters**, creates a truncated distributed towards shallow waters.

Biodiversity data cleaning

Particularly important for data sourced from warehouses such as GBIF.

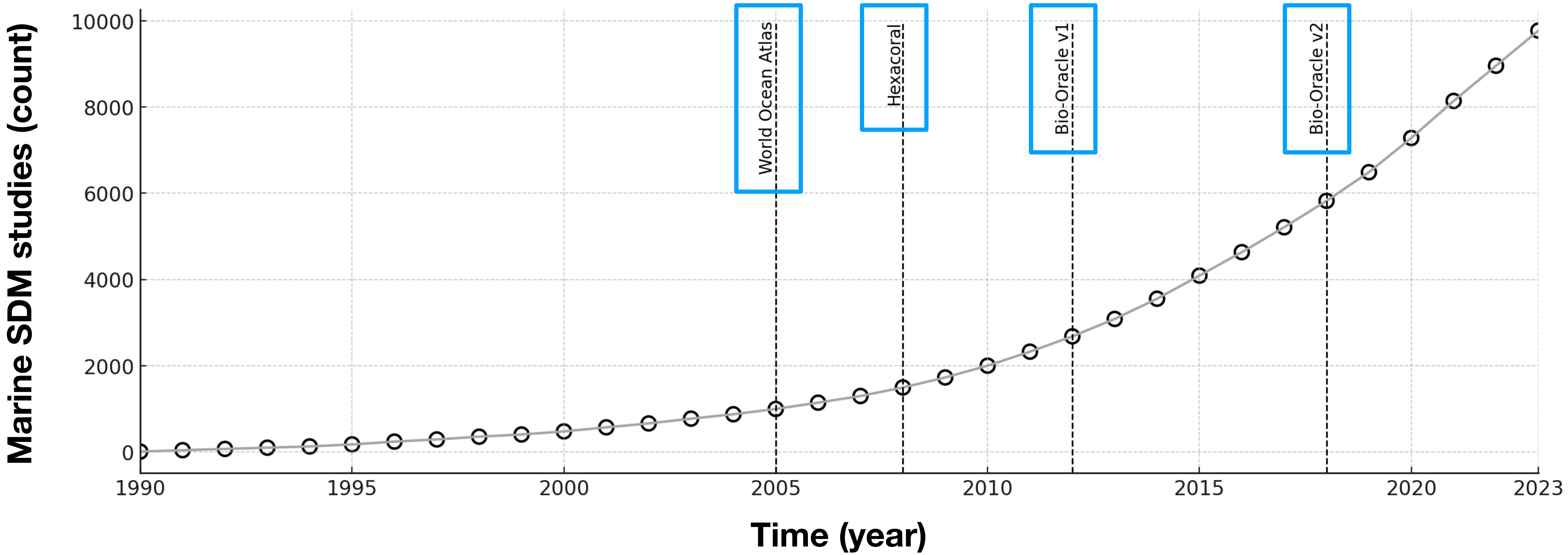
1. Remove records with NA coordinates;
2. Remove records on land or outside known distributions (depth also);
3. Remove duplicate records.



Environmental data

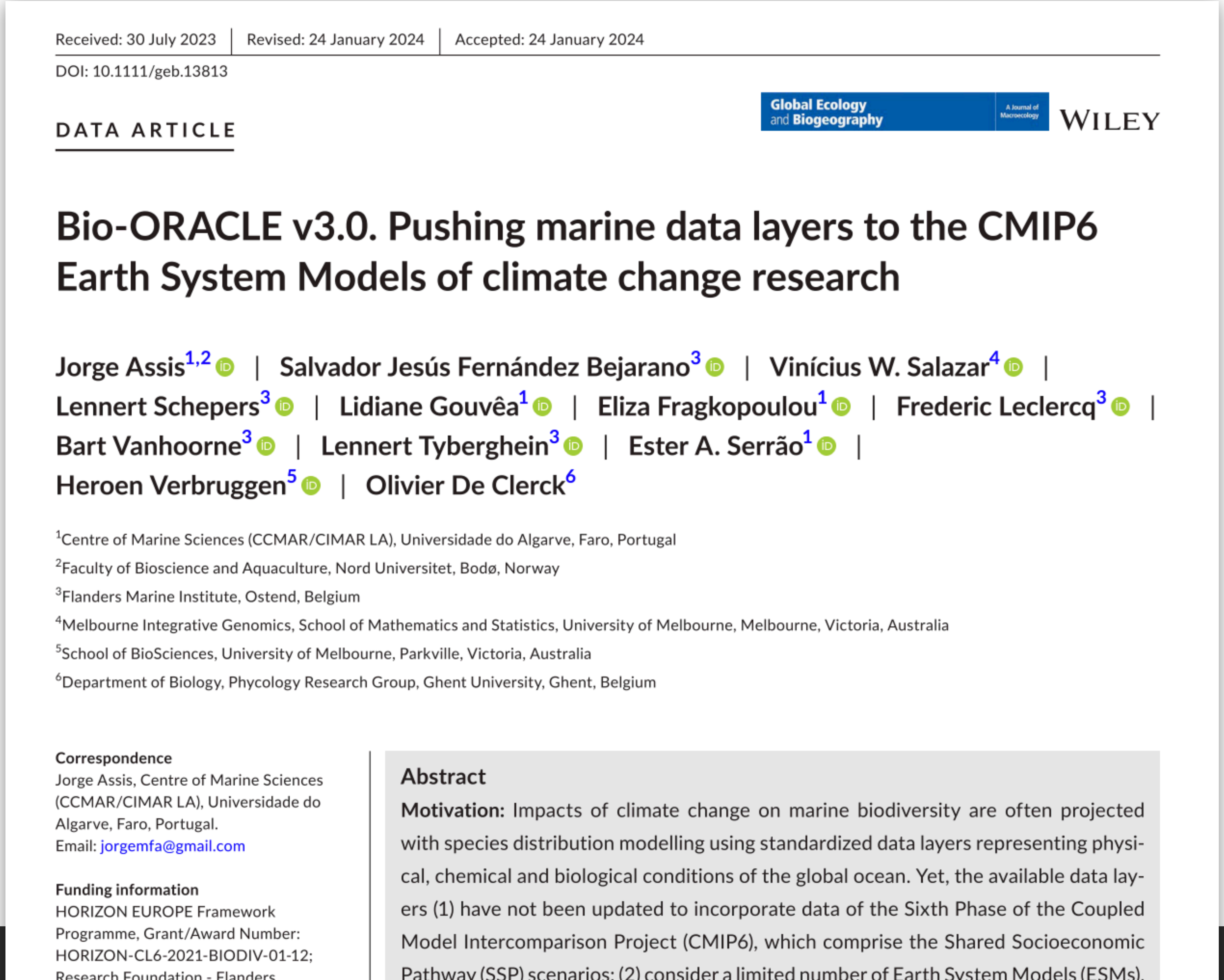
Allows **linking biodiversity data with their extant conditions**.

Environmental predictors typically in the form of digital raster layers.



Increase use of marine SDM in climate change projections owing to pioneer initiatives ** that provide **open-source GIS data layers with uniform resolution, global coverage**, and variables with **biological significance**.

biodiversityDS.



Received: 30 July 2023 | Revised: 24 January 2024 | Accepted: 24 January 2024
DOI: 10.1111/geb.13813

DATA ARTICLE

Global Ecology and Biogeography — A Journal of Macroecology
WILEY

Bio-ORACLE v3.0. Pushing marine data layers to the CMIP6 Earth System Models of climate change research

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Funding information
HORIZON EUROPE Framework Programme, Grant/Award Number: HORIZON-CL6-2021-BIODIV-01-12; Research Foundation - Flanders

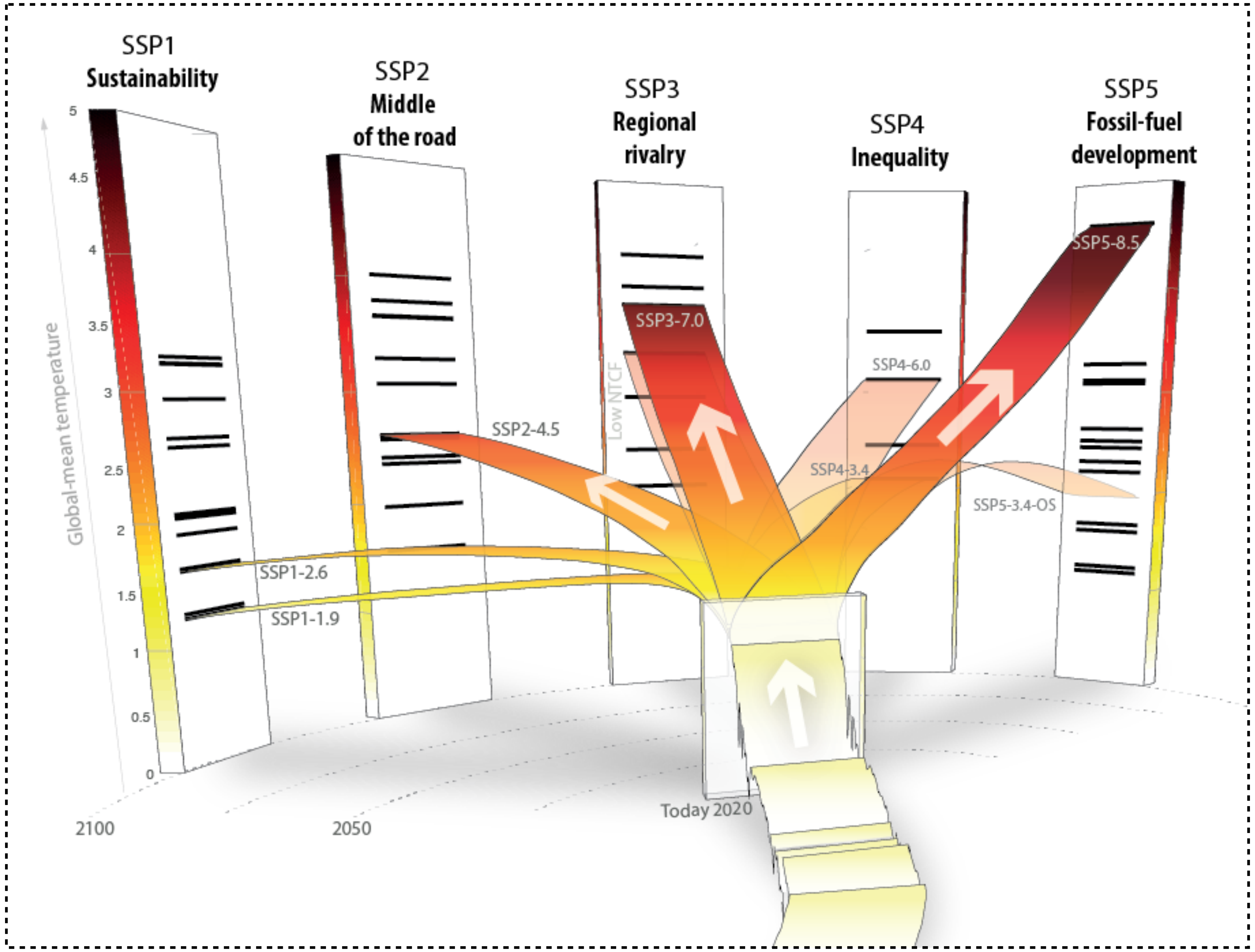
Abstract

Motivation: Impacts of climate change on marine biodiversity are often projected with species distribution modelling using standardized data layers representing physical, chemical and biological conditions of the global ocean. Yet, the available data layers (1) have not been updated to incorporate data of the Sixth Phase of the Coupled Model Intercomparison Project (CMIP6), which comprise the Shared Socioeconomic Pathway (SSP) scenarios; (2) consider a limited number of Earth System Models (ESMs),

Bio-Oracle improvements

Layers for future conditions **consider data from the 6th phase of the Coupled Model Inter-comparison Project.**

CMIP6 (1) is **more accurate in capturing the spatial and temporal variability of oceanographic conditions** and (2) **outputs the SSP scenarios**, which are the new narratives of global change.



The new **SSP use up-to-date socioeconomic assumptions** and **encompass the Paris Agreement expectations**, allowing the development of **projections aligned with international climate policies**.

Bio-Oracle improvements

Layers for future conditions comprise numerous Earth System Models.

ESM are simulations that resolve global atmospheric and oceanographic conditions across space and time (e.g., end-of-century SST in Culatra island).

ESM have inherent uncertainties.

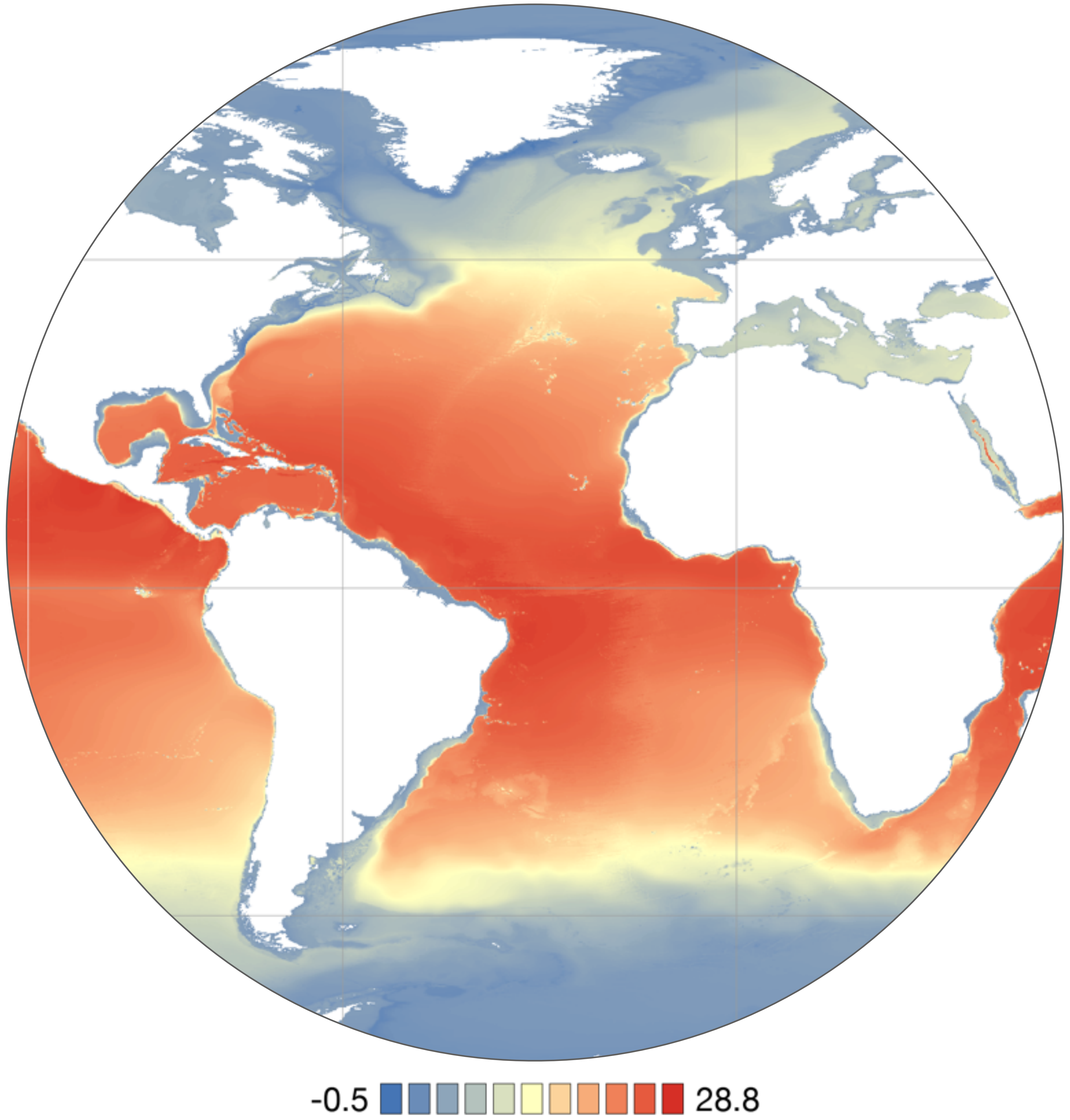
The multi-model ensemble (e.g., averaging different models) to account for inter-ESM variability. Previous available datasets limited to 2 ESMs (from the dozens available), limiting the potential to capture the range of climate change.

Bio-Oracle improvements

Layers for future conditions consider biologically meaningful variables beyond temperature, salinity and sea ice conditions.

Oxygen, primary productivity and pH (among others) are limiting factors for biodiversity, and are expected to change in the future.

Failing to consider these variables in climate change projections might misinterpret the exposure of biodiversity to future detrimental conditions.

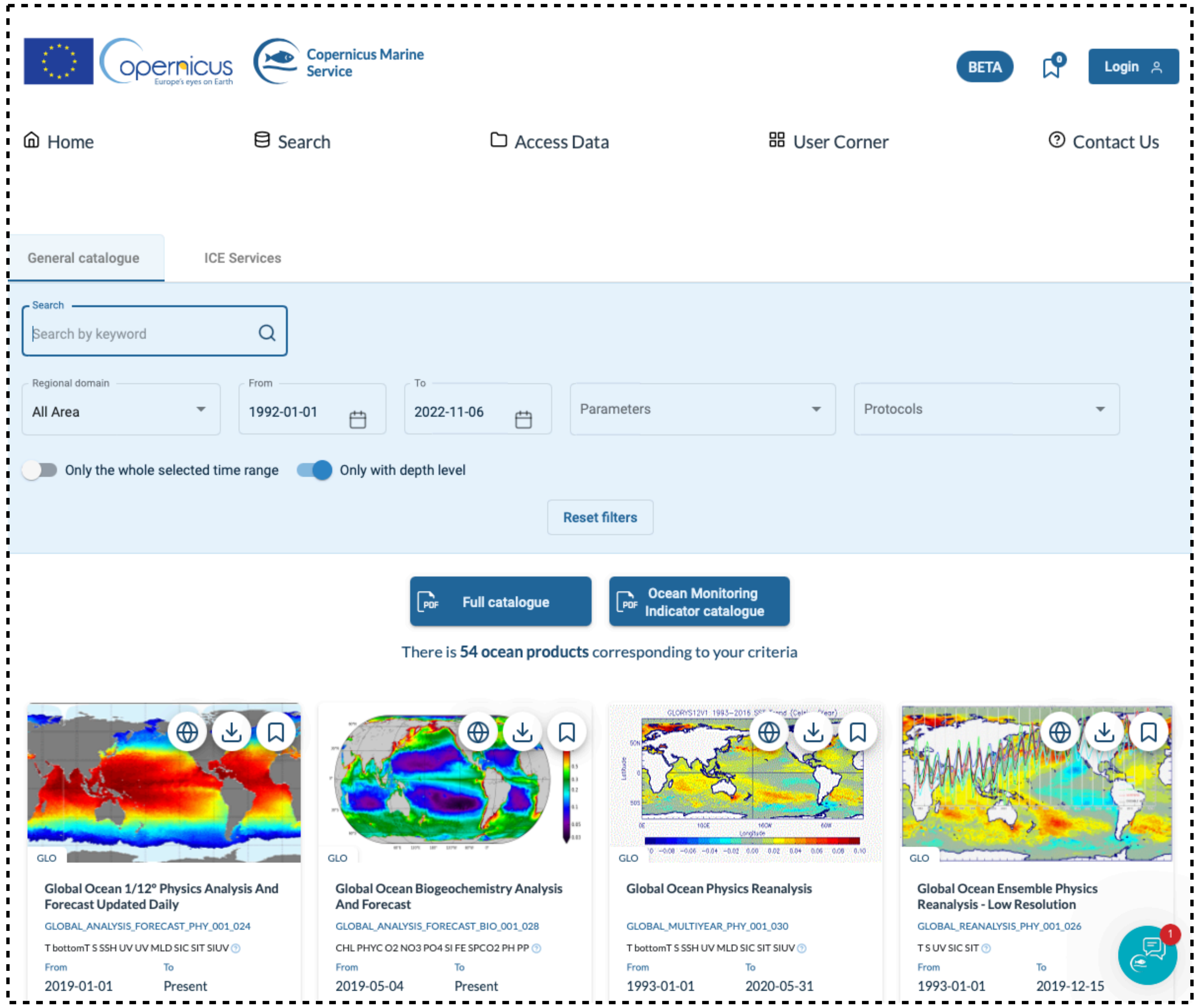


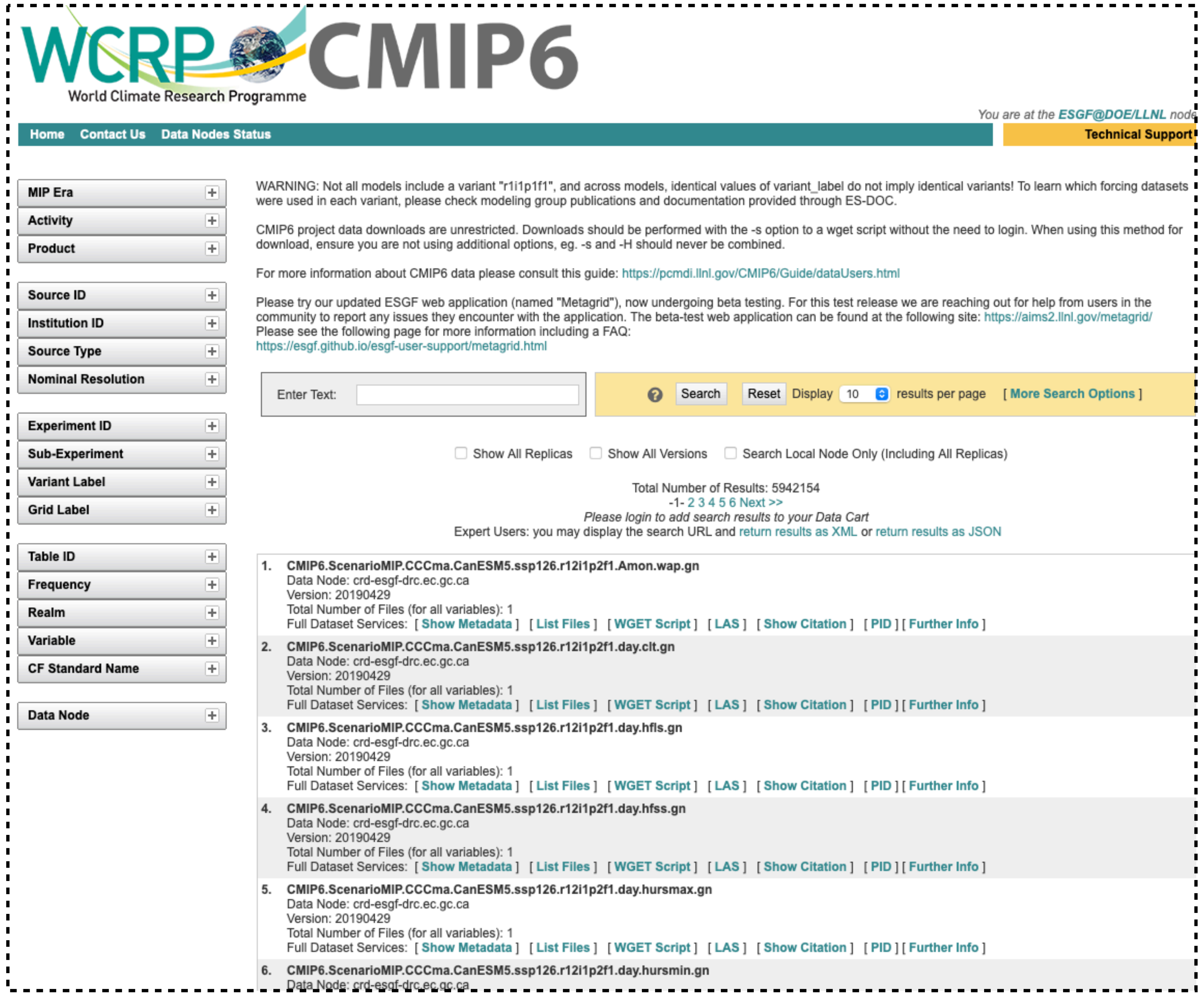
Difference between surface and bottom temperature

All layers also provided for bottom conditions, to better **model and project changes of benthic species** (e.g., deep corals, or demersal fish).

		Present	SSP1-1.9	SSP1-2.9	SSP2-4.5	SSP3-7.0	SSP4-6.0	SSP5-8.5
		2 (2000-2020)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)
1	**Temperature**							
2	**Salinity**							
3	**Sea Ice Cover**							
4	**Sea Ice Thickness**							
5	**Sea Water Velocity**							
6	**Mixed Layer Depth**							
7	**Diffuse Attenuation Coefficient**							
8	**PAR**							
9	**PAR at Bottom**							
10	**Oxygen**							
11	**pH**							
12	**Iron**							
13	**Phosphate**							
14	**Nitrate**							
15	**Silicate**							
16	**Primary Production**							
17	**Phytoplankton**							
18	**Chlorophyll**							

Extension of Bio-ORACLE: **18 essential variables** provided for the surface and along the seafloor, **from 2000 to 2100**, **per decade**, at **0.05º resolution**, **covering the span of SSPs**, by **ensembling 10 ESMs of CMIP6**.





Data from Copernicus Marine Service and World Climate Research Programme

Copernicus
Europe's eyes on Earth
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Search by keyword
Regional domain
All Area
GLO
Forecast Updated Daily
From
To
2019-01-01
Present
WCRP
CMIP6
World Climate Research Programme
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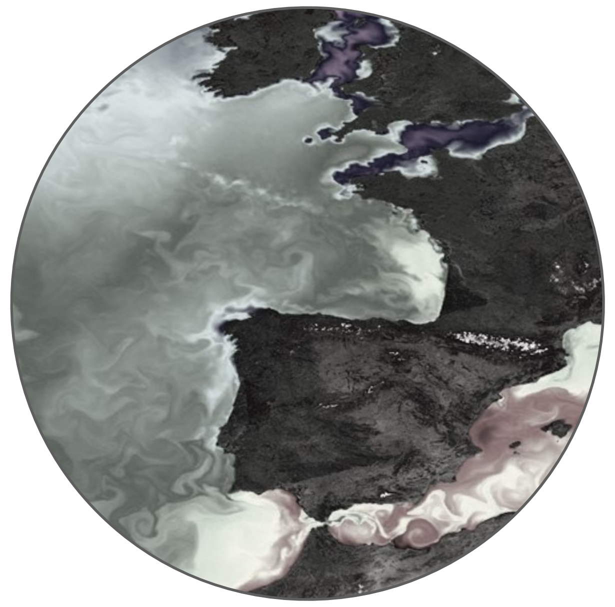
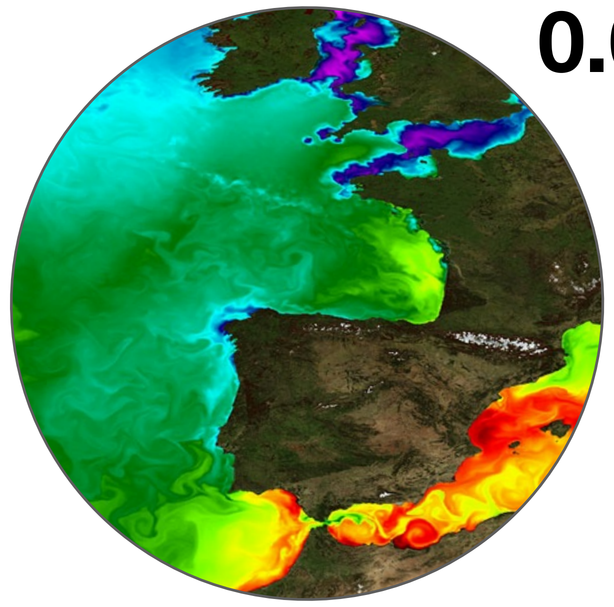
Technical Support
This helps evaluate climate models, leads to improvements in the model simulations and provides a better understanding of past, present and future climates.

Data f **me**



Copernicus
1993-2023

Subsetting data and
resampling to BO resolution



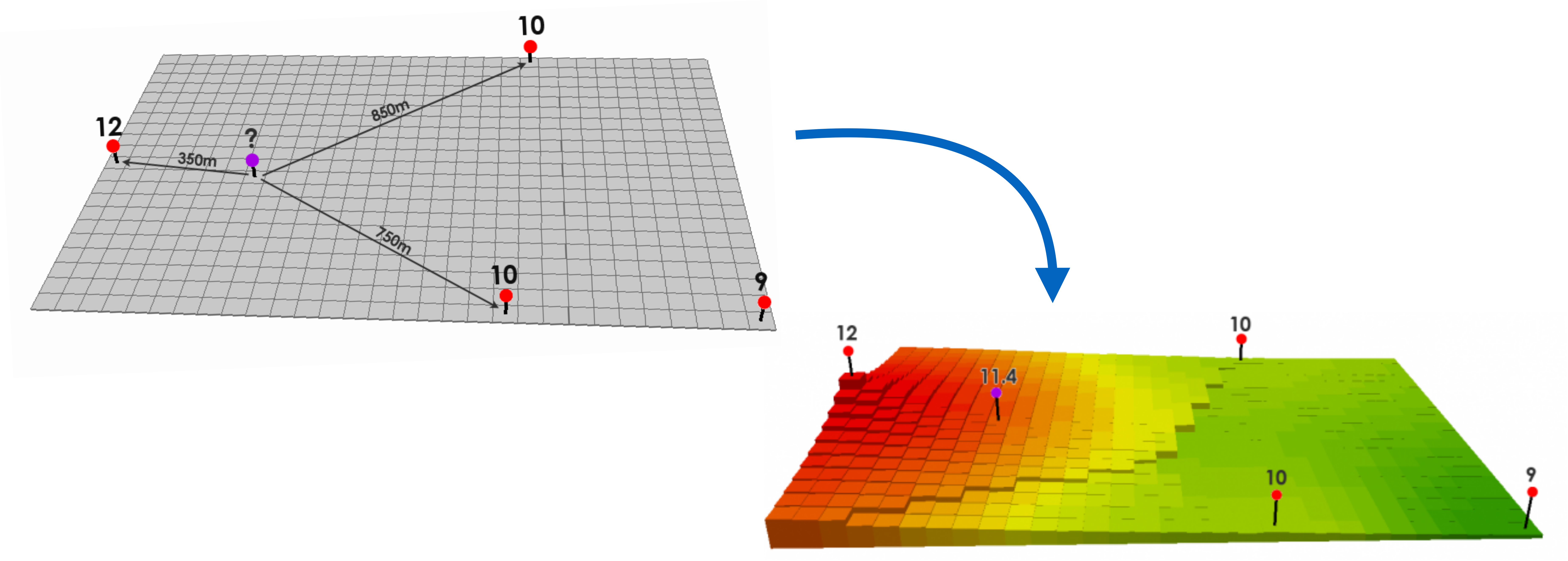
0.05º

BO Layer
2010-2020

For present-day conditions, layers are developed by subsetting data to the **desired decade** and **resampling** (interpolating) to the resolution of BO (0.05º).

e.g.,

layer 1: Global maximum sea surface temperature of 2010-2020



Interpolation

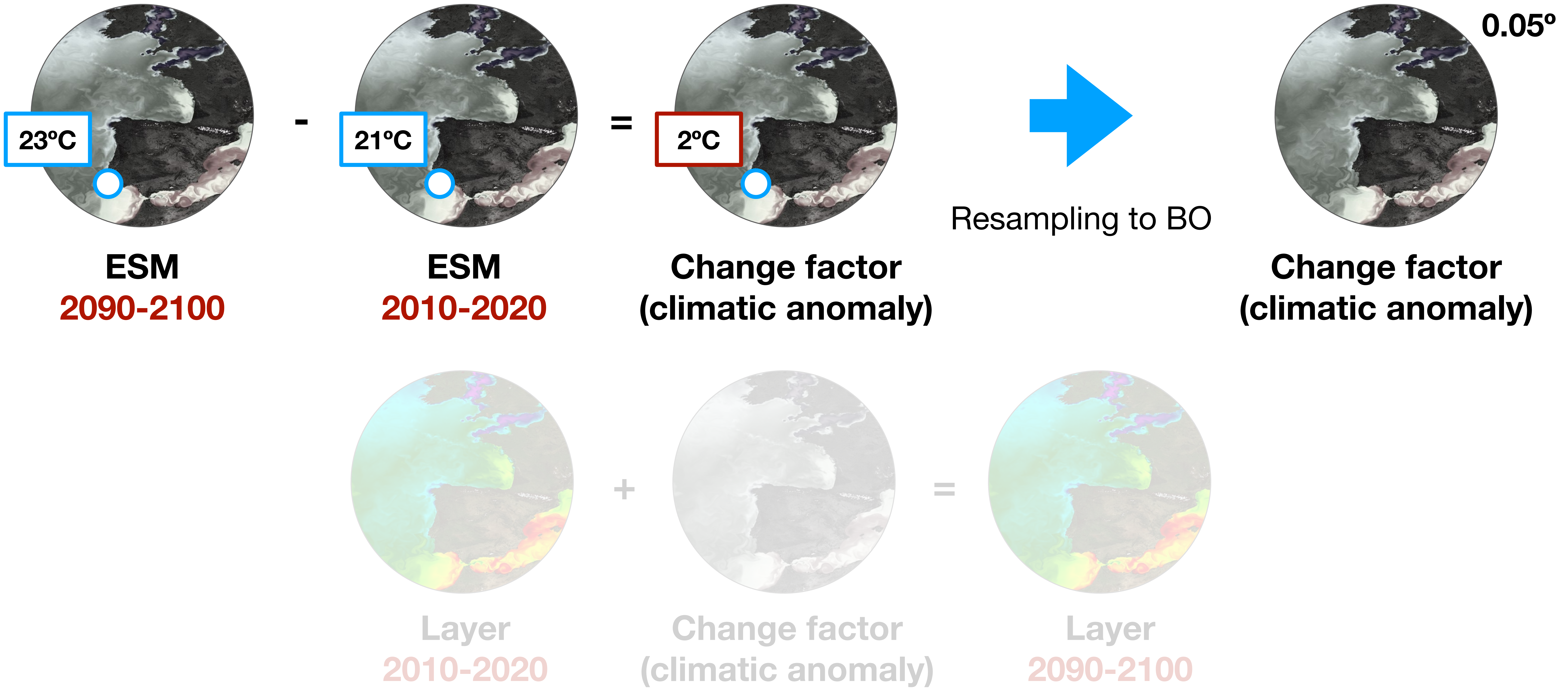
Data are collected from numerous sources (e.g., buoys, loggers in the field) and their information are spatially interpolated to infer the distribution of the environmental gradients for the whole region - **estimates unknown points**.

e.g., Inverse Distance Weighting algorithm.

For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

layer 2: Global maximum sea surface temperature of 2090-2100



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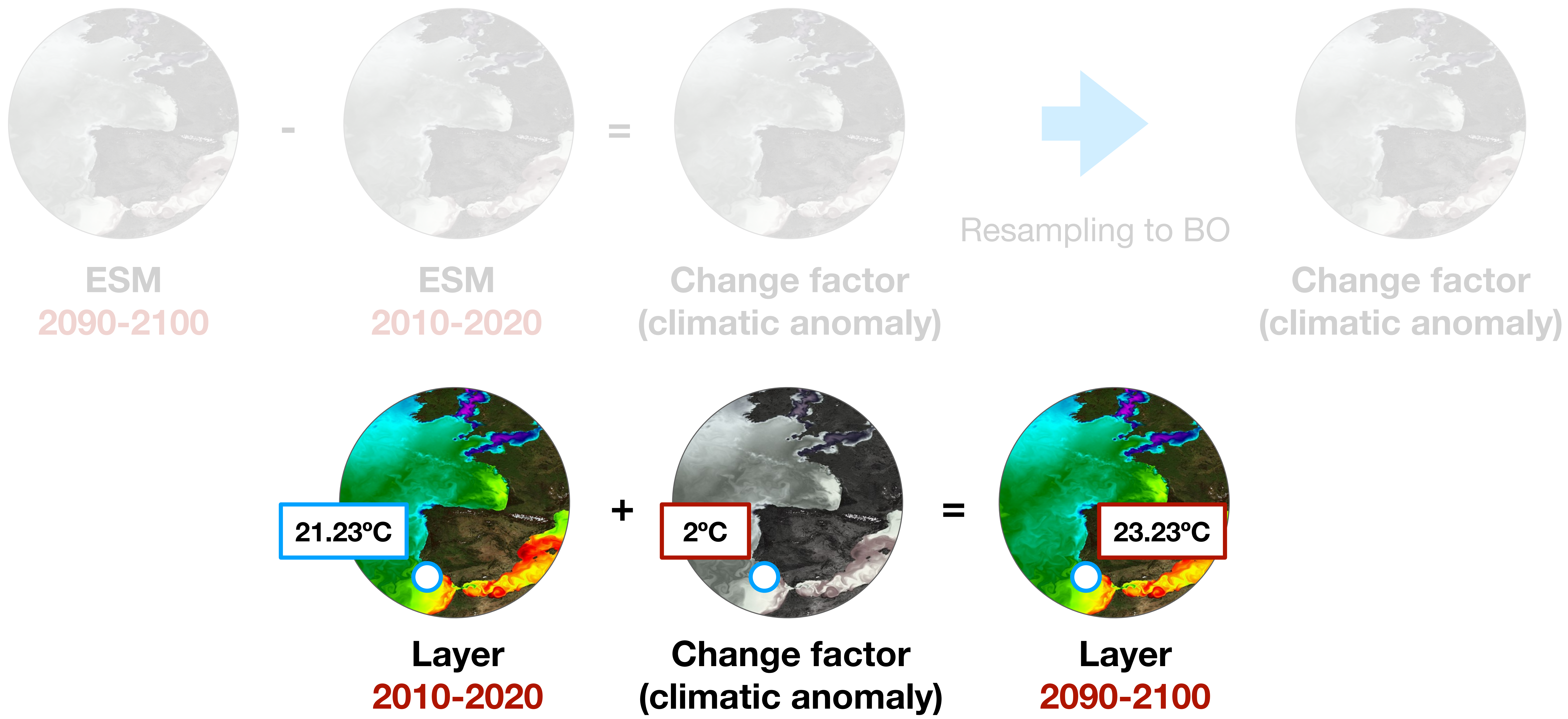

For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

layer 2: Global maximum sea surface temperature of 2090-2100

The layers are provided under the FAIR principle at:

(1) An ERDDAP instance at the Flanders Marine Institute Data Centre **

http://erddap.bio-oracle.org

(2) Bio-ORACLE website

http://www.bio-oracle.org

(3) **R and Python packages** for programmatic data access and easy integration in available SDM workflows.

Code used to generate the layers available in the permanent GitHub repository at **https://github.com/bio-oracle/data-layers**.